HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

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SLIIT
FACULTY OF COMPUTING

Introduction Video

HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

Problem & Its Impact/Gravity

- **Manual Microscopy:** Traditional peripheral blood smear examination by pathologists
- **Time Delay:** 3-7 days for results in government hospitals due to specimen transport and specialist availability
- **Human Error:** Inter-observer variability 20-35%, subjective assessment prone to misdiagnosis
- **Rural Access:** 80% of rural clinics lack hematology specialists

Who Will Be Benefited

- **Patients:** early diagnosis, reduced mortality
- **Healthcare Professionals (Lab technicians, Hematologists):** time reduction of disease identification and prediction

Design Excellence

- **Multi-modal approach:** Combines blood smear image analysis + CBC parameters
- **Unified platform:** Identifying Four diseases (ALL, Thalassemia, IDA, SCD) in one system
- **Hybrid models:** Combined technologies to identify the diseases
- **Modern Technology:** Django framework, React, Tailwind CSS

Risk Mitigation

- Due to delay of ethical clearance documents, we had to start the model developments using a common dataset until we get the data access approval from hospitals and labs
- Data Augmentation is used for small datasets.

IT22371218 | Aluthge A.D.M.S

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

Specialization: Software Engineering

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

Problem Definition

- Acute Lymphoblastic Leukemia (ALL) diagnosis in Sri Lanka has 3-7 day delays in test results due to and shortage of specialist hematologists and limited laboratory facilities
- Traditional manual microscopic examination is error-prone and relies on doctor's subjective assessment

Proven Gap

- Current research approaches ALL detection through isolated methodologies - either blood smear image analysis OR CBC parameter analysis but lacks integration of both modalities
- No published work on integrated hybrid systems for ALL detection tailored to Sri Lankan patient populations, with most international models trained on Western datasets

Solution Approach

- **Hybrid models:** Combined technologies to identify Acute Lymphoblastic Leukemia (ALL)
- **Multi-modal approach:** Combines both blood smear images and CBC reports to come to a prediction
- **Sri Lankan Dataset:** Sri Lankan dataset is used to train models and extract CBC parameters

How Problem Will Be Addressed

- Manual microscopic examination is replaced by AI models and it will make the diagnosis faster and reduce the human error
- Since ALL is related to genes, using a Sri Lankan dataset to train makes the predictions better
- Using multi-model approach improves the accuracy of the predictions

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

User Requirements

- **Lab Technician** - Easy image upload and disease identification
- **Doctor** - Reliable diagnosis + confidence
- **Patient** - Quick results

Design Excellence

- **Simple medical-grade UI**
- **Multi-modal approach:** Combines blood smear image analysis + CBC parameters
- **Hybrid models:** Combined technologies to identify the diseases
- **Modern Technology:** Django framework, React, Tailwind CSS

User Feedback on Prototype

- Will be presented in common

Functional Requirements

- Upload blood smear image
- Upload CBC report
- Real-time processing
- Diagnosis output
- Automated lymphoblast detection

Contribution

- Create figma designs and react components for the login, lab technician dashboard, all patients page for lab technician, add patient function, edit patient function, view patient information function and delete patient function.
- Created role based authentication using JWT and integrated with frontend
- Created base models with CNN, ResNet50, EfficientNetB0, YOLOv8s, YOLOv8n, YOLO11 and combined CNN and EfficientNetB0 to make one hybrid model and combined YOLOv8s and CNN to make another hybrid model

IT22368348 | Disanayake R.G.O.S

Detection and Identification of Beta-Thalassemia (Minor)

Specialization: Software Engineering

Detection and Identification of Beta-Thalassemia (Minor)

Problem Definition

- Current diagnosis relies on manual microscopy, which are costly, time-consuming, and not widely accessible, while no Sri Lanka–specific AI diagnostic system currently exists.
- Misdiagnosis risk: Beta-Thalassemia Minor vs. Similar diseases due to overlapping CBC indices.

Proven Gap

- There is no hybrid AI model that combines CBC data with blood smear images, and most existing systems are trained on foreign datasets, making them unsuitable for Sri Lankan patients.
- Furthermore, current methods struggle to differentiate Beta-Thalassemia Minor from similar disease and lack a web-based platform for real-time clinical deployment.

Solution Approach

- **Hybrid models:** Combined technologies to identify Beta -Thalassemia (Minor)
- **Multi-modal approach:** Combines both blood smear images and CBC reports to come to a prediction
- **Sri Lankan Dataset:** Sri Lankan dataset is used to train models and extract CBC parameters

How Problem Will Be Addressed

- Manual microscopic examination is replaced by AI models and it will make the diagnosis faster and reduce the human error
- Since Beta Thalassemiais related to genes, using a Sri Lankan dataset to train makes the predictions better
- Using multi-model approach improves the accuracy of the predictions

Detection and Identification of Beta-Thalassemia (Minor)

User Requirements

- **Lab Technician** - Easy image upload and disease identification
- **Doctor** - Reliable diagnosis + Confidence + Feedbacks for results
- **Patient** - Quick results

Functional Requirements

- Upload blood smear image
- Upload CBC report
- Real-time processing
- Diagnosis output
- Automated morphology detection

Design Excellence

- **Simple medical-grade UI**
- **Multi-modal approach:** Combines blood smear image analysis + CBC parameters
- **Hybrid models:** Combined technologies to identify the diseases
- **Modern Technology:** Django framework, React, Tailwind CSS

Contribution

- Create figma designs and react components for the doctor dashboard, all patients view for doctor , view diagnosis results for doctor and view diagnosis results for lab technician , Calender , admin dashboard
- Initiated User CRUD for user details managment function.
- Created base models with CNN, ResNet50, EfficientNetB0, YOLOv8, YOLO11
- Create ResNet50 + EfficientNet and YOLOv8 + ResNet50 hybrid models

User Feedback on Prototype

- Will be presented in common

IT22361936 | Dassanayake E.G.N.L

Detection and Identification of Sickle Cell Disease (SCD)

Specialization: Software Engineering

Detection and Identification of Sickle Cell Disease (SCD)

Problem Definition

- Lack of laboratory facilities and the specialized hematologists delay the Sickle Cell Disease (SCD) diagnosis.
- Misdiagnosis risk: Mostly appear as trait untill it meets the extreme conditions.

Proven Gap

- There is no hybrid AI model that combines CBC data with blood smear images, and most existing systems are trained on foreign datasets, making them unsuitable for Sri Lankan patients.
- Existing approaches struggles to identify the Sickle Cell Disease (SCD) clearly. No proper software solution has developed for this yet.

Solution Approach

- **Hybrid models:** Combined technologies to identify Sickle Cell Disease (SCD)
- **Multi-modal approach:** Combines both blood smear images and CBC reports to come to a prediction
- **Sri Lankan Dataset:** Sri Lankan dataset is used to train models and extract CBC parameters

How Problem Will Be Addressed

- Manual microscopic examination is replaced by AI models and it will make the diagnosis faster and reduce the human error
- Sickle Cell Disease is gene related disease, therefore the best option is to use a Sri Lankan dataset.
- Using multi-model approach improves the accuracy of the predictions

Detection and Identification of Sickle Cell Disease (SCD)

User Requirements

- **Lab Technician** - Easy image upload and disease identification
- **Doctor** - Reliable diagnosis + Confidence + Feedbacks for results
- **Patient** - Quick results

Design Excellence

- **Simple medical-grade UI**
- **Multi-modal approach:** Combines blood smear image analysis + CBC parameters
- **Hybrid models:** Combined technologies to identify the diseases
- **Modern Technology:** Django framework, React, Tailwind CSS

User Feedback on Prototype

- Will be presented in common

Functional Requirements

- Upload blood smear image
- Upload CBC report
- Real-time processing
- Diagnosis output
- Automated morphology detection

Contribution

- Create figma designs and react components for the create, view, edit, remove functionalities for every user(admin, lab technician, doctor.
- Setup the initial project structure with django framework with containerization using docker.
- Created the user creation, view, update, delete functionalities for every user role.
- Created base models with ResNet50, EfficientNetB0. The R-CNN and YOLO11 models are underdeveloped

IT22326140 | Sachinthanee M.A.S

Detection and Identification of Iron Deficiency Anemia (IDA)

Specialization: Software Engineering

Detection and Identification of Iron Deficiency Anemia (IDA)

Problem Definition

- Iron Deficiency Anemia is the most common form of anemia in Sri Lanka, but diagnosis is still performed manually using CBC parameters or blood smear analysis, resulting in slow and specialist-dependent workflows.
- Misdiagnosis risk: Iron Deficiency Anemia vs. Similar diseases due to overlapping CBC indices.

Proven Gap

- There is no hybrid AI model that combines CBC data with blood smear images, and most existing systems are trained on foreign datasets, making them unsuitable for Sri Lankan patients.
- Current diagnostic methods face challenges in distinguishing Iron Deficiency Anemia from related diseases and are not supported by real-time web-based clinical systems.

Solution Approach

- **Hybrid models:** Combined technologies to identify Iron Deficiency Anemia
- **Multi-modal approach:** Combines both blood smear images and CBC reports to come to a prediction
- **Sri Lankan Dataset:** Sri Lankan dataset is used to train models and extract CBC parameters

How Problem Will Be Addressed

- Manual microscopic examination is replaced by AI models and it will make the diagnosis faster and reduce the human error
- Using multi-model approach improves the accuracy of the predictions

Detection and Identification of Iron Deficiency Anemia (IDA)

User Requirements

- **Lab Technician** - Easy image upload and disease identification
- **Doctor** - Reliable diagnosis + Confidence + Feedbacks for results
- **Patient** - Quick results

Functional Requirements

- Upload blood smear image
- Upload CBC report
- Real-time processing
- Diagnosis output
- Automated morphology detection

Design Excellence

- **Simple medical-grade UI**
- **Multi-modal approach:** Combines blood smear image analysis + CBC parameters
- **Hybrid models:** Combined technologies to identify the diseases
- **Modern Technology:** Django framework, React, Tailwind CSS

Contribution

- Create figma designs and react components for profile view for logged in user (Doctor, Lab Technician, Admin) edit logged in user profile (Doctor, Lab Technician, Admin), Change password for logged in user (Doctor, Lab Technician, Admin)
- Change password function.
- Created base models with CNN, ResNet50, EfficientNetB3, DenseNet121,yolo v8 CLS ,yolo v11
- Hybrid models:DenseNet121+efficientNetB3

User Feedback on Prototype

- Will be presented in common

HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

Customer Persona

- **Customers:** Hospitals, diagnostic labs, NGOs, rural clinics
- **Users:** Hematologists, lab technicians, physicians

Market Size

- **Growing Digital Healthcare :** More hospitals adopt digital systems.
- **High Blood Disorder Prevalence :** Common among local patients.
- **Limited Specialists :** Few experts available.
- **High Patient Load** : Overcrowded public hospitals.

What Makes It Unique

- **Hybrid AI Model :** Uses multiple technologies to identify blood disorders.
- **Multi-Model Analysis :** Combines blood smear images and CBC reports for prediction.
- **Sri Lankan Dataset :** Models trained using local blood data and CBC parameters.

Cost Recovery

- Subscription for hospitals/labs (e.g., LKR 10,000 – 15,000 / month)
- Pay-per-use for clinics (e.g., LKR 200 – 300 / test)
- Possible government/NGO funding for scaling